

1.4 Requirements Specification

Project: E-Learning Web Pages for Computer Science

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1. Purpose

The purpose of this section is to define the **functional** and **non-functional** requirements of the e-learning system being developed.

The project aims to design and implement a **web-based CPU pipeline simulator** that enables students to visualise, interact with, and understand instruction pipelining concepts such as hazards, forwarding, and performance metrics.

Requirements are derived from project objectives, pedagogical goals, and discussions with the supervisor.

2. Overview

The system will allow users to:

- Input or select assembly-like instructions.
 - Observe how instructions move through the **five pipeline stages** (Fetch, Decode, Execute, Memory, Write-back).
 - Identify and understand different **pipeline hazards** (data, structural, control).
 - Experiment with **hazard resolution techniques** such as stalling and forwarding.
 - View and interpret **performance statistics** (cycles, CPI, branch prediction accuracy).
 - Complete an **anonymous SLEA questionnaire** to evaluate the educational effectiveness and usability of the tool.
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3. Functional Requirements

ID	Requirement Description	Priority (MoSCoW)	Testable Criteria
FR1	The system shall allow users to input assembly-like code using a built-in text editor with syntax highlighting.	Must	Verify input is accepted and parsed without errors.
FR2	The simulator shall visually display the flow of each instruction through the 5 pipeline stages (IF, ID, EX, MEM, WB).	Must	Confirm stage transitions occur per clock cycle.
FR3	The system shall detect and highlight data hazards (RAW, WAR, WAW).	Must	Trigger hazard warnings when dependent instructions appear.
FR4	The system shall detect and visualise control hazards caused by branch instructions.	Must	Display branch prediction and misprediction cycles.
FR5	The simulator shall demonstrate hazard resolution techniques (stalling, forwarding, branch prediction).	Must	Observe correct bubble insertion or forwarding arrows.
FR6	The system shall include a play, pause, and step control for simulation speed.	Should	Verify responsive user controls and accurate time steps.
FR7	The system shall display register and memory values in real-time.	Must	Compare values with expected CPU state after execution.
FR8	The system shall calculate and display performance metrics (CPI, total cycles, throughput).	Should	Confirm correct numerical results after simulation.
FR9	The system shall provide guided examples and tooltips to support student learning.	Should	Display tooltips when hovering over pipeline stages.
FR10	The system shall include an SLEA questionnaire to evaluate user learning experience anonymously.	Must	Confirm responses are recorded anonymously and stored securely.
FR11	The system shall allow users to reset and re-run simulations.	Must	Verify state resets correctly without page reload.
FR12	The simulator shall be deployable and accessible via a public URL.	Must	Access application via modern web browsers.

4. Non-Functional Requirements

ID	Requirement Description	Priority (MoSCoW)	Testable Criteria
NFR 1	The simulator shall be fully browser-based (no installation).	Must	Runs on Chrome, Firefox, and Edge without setup.
NFR 2	The user interface shall be responsive across screen sizes.	Should	Works on desktop and laptop displays.
NFR 3	The application shall comply with WCAG 2.1 Level AA accessibility standards.	Must	Passes automated accessibility validation.
NFR 4	The system shall maintain performance of ≥ 30 FPS for animations.	Should	No visible lag during normal operation.
NFR 5	The system shall load completely within 3 seconds on a 10 Mbps connection.	Should	Measured using browser DevTools.
NFR 6	The codebase shall be modular and follow MVC architecture.	Must	Verified through inspection and code documentation.
NFR 7	The system shall use open-source libraries under approved licences (MIT, Apache 2.0).	Must	Check all dependencies before deployment.
NFR 8	The system shall store all SLEA responses anonymously and locally (no personal data).	Must	Ensure GDPR compliance.
NFR 9	The system shall support multiple simultaneous users (static hosting).	Could	Verify no conflicts under concurrent sessions.

5. External Interface Requirements

Interface Type	Description
User Interface	Browser-based graphical layout showing pipeline stages, control buttons, and data panels. Includes input text editor and performance metrics display.

Hardware Interface	None required — system runs on standard computing devices.
Software Interface	Uses browser APIs (Canvas, localStorage) and optional libraries (React, Chart.js).
Communication Interface	No backend server; static deployment via HTTP/HTTPS.

6. Traceability Matrix

Objective / Aim	Related Requirement IDs
Visualise 5-stage pipeline operation	FR2, FR3, FR4, FR5
Detect and explain hazards	FR3, FR4, FR5, FR9
Measure and display performance	FR7, FR8
Support interactive learning	FR1, FR6, FR9
Evaluate user understanding (SLEA)	FR10, NFR8
Ensure accessibility and portability	NFR1, NFR2, NFR3

7. Summary

The requirements defined above provide a foundation for developing a **functional, educational, and accessible CPU pipeline simulator**.

They ensure the project meets both **technical** goals (performance, compatibility, maintainability) and **pedagogical** objectives (interactivity, active learning, feedback collection).

All requirements will be validated during system testing and user evaluation through the SLEA questionnaire.